

IANS

FACE OF THE FUTURE A conceptual rendering of India's proposed bullet train on display at Rail Bhavan, New Delhi

SIGNALLING

India has opted for the European Train Control System (ETCS) Level 2, replacing the Japanese Shinkansen's proprietary DS-ATC system for its Mumbai-Ahmedabad HSR project

- **A globally used standard, ETCS is interoperable across rail networks**
- **Multiple suppliers can compete, lowering costs**
- **Relies on radio communication instead of DS-ATC's copper-heavy cable network, reducing theft risk**

CIVIL WORKS

India bargained hard to break from the older Japanese funding model, under which Japanese companies led execution while Indian firms largely worked as junior partners

- **L&T is the principal civil contractor, building most of the elevated corridor, stations, depots, tracks etc.**
- **Afcons is building the 21-km underground tunnel and the MEIL-HCC joint venture the underground terminus, boosting domestic expertise**

Little wonder that when the Indian and Japanese premiers came together in Ahmedabad on September 14, 2017, PM Modi made no attempt to hide his gratitude. "The credit for this bullet train goes entirely to my dear friend Shinzo Abe," he said at the groundbreaking ceremony. "*Yeh bullet train ek prakaar se Japan ka Bharat ko bahut bada saugaat hai...ek prakaar se muft mein yeh poora project banta chala ja raha hai* (This bullet train is Japan's gift to India. In a way, the project is being built almost for free)."

Japanese generosity, however, came accompanied by a precondition. The project's core systems—from trains and signalling to electrical equipment—were to be sourced from Japanese companies or consortia led by them, leaving India with little flexibility. It soon became evident that bids from a limited pool of Japanese vendors were significantly costlier than international alternatives. As costs escalated and delays mounted, the political optics became increasingly difficult to ignore. It was then that India realised it had to gradually untangle itself from parts of the original procurement framework.

THE FIRST BREAKTHROUGH

Civil construction was the first to break out of Japan's exclusivity clause. It involved prolonged negotiations, what officials called "rail diplomacy". "We had to bargain hard with Japan," recalls Mukul Saran Mathur, former general manager of Indian Railways, who steered the project in its initial phase. "They eventually relented, but on the condition that there would be strict Japanese oversight on the quality of the work, as India had never constructed anything like this before." But Japanese firms executing even the civil works would be "needlessly expensive", according to Mathur. "Besides, how would our engineering capability upgrade if we remain just a consumer of the technology and not a builder?" The waiver of that requirement opened the door for Indian companies such as Afcons and L&T to take charge of large parts of the civil works.

But building tracks, bridges and stations was only one part of creating an HSR network. Two critical pieces remained—the trains themselves and the signalling system. It was here that India ran into a far bigger hurdle.